

The Current MES-Bound Framework and Planck Low-Multipole Results

A self-contained account of the theory, finite-pool statistics,
scalar mechanism audit, observer-space irreducible-tensor diagnostics,
and registered carrier-domain method power

Referee-repaired second revision

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Abstract

This report gives a self-contained statement of the current Maartens–Ellis–Stoeger (MES) bound framework and all presently admitted Planck low-multipole results obtained after the statistical-method upgrade. The theory is formulated as a one-way, premise-conditioned ceiling on the squared Hubble-normalized shear and vorticity invariants. On the registered geodesic, residual-dipole-free branch, the two ceilings are deterministic functions of the measured quadrupole and octupole powers; the transformation is invertible on its nonnegative domain and therefore creates neither a sky direction nor independent information. Direction and shape enter only through the actual direction-indexed harmonic field, represented here by real $\ell = 2, 3$ symmetric trace-free (STF) tensors.

The statistical analysis treats one Planck PR3 SMICA observation and finite FFP10 null ensembles symmetrically. In the exact paired 300-row CMB-plus-noise pool, the squared MES ceilings are $\Sigma_{\max}^2 = 2.4000870 \times 10^{-10}$ and $W_{\max}^2 = 3.0061888 \times 10^{-13}$, with local ranks 74/301 each. The ten-coordinate MES-plus-morphology family has rank 98/301 under the legacy leave-one-out absolute-median reducer; an all-two-sided factorial audit gives 148/301 under the leave-one-out ECDF-midrank reducer, while a separately predeclared physics-oriented tail registry gives 93/301. In the observer-space STF analysis, the eight-coordinate frame-free family has rank 27/301 under both reducers in the paired CMB-plus-noise pool. The same frozen observation and feature family give 61/1000 and 55/1000 in a separate 999-row CMB-only pool. This change is a descriptive null-ensemble sensitivity, not an independent confirmation. An exact observation-row replacement audit, with the paired 300-row null pool and every statistic definition held fixed, gives frame-free ranks 27/301, 46/301, 27/301, and 81/301 for SMICA, Commander, NILC, and SEVEM under the absolute-median reducer; the ECDF ranks are 27/301, 41/301, 27/301, and 40/301. WMAP9 ILC, CosmoGlobe, and BeyondPlanck CMB holdouts were processed by the same operator. A predeclared 13-template foreground bank gives a simultaneous Q/O projection rank of 144/301: SMICA is not exceptionally aligned with a single admitted template under this necessary-condition diagnostic.

A separate, observation-free intervention study now measures the method power of the frozen MES-10 and frame-free-irrep families. It injects five unit-energy algebraic $\ell = 2, 3$ templates, at seven registered amplitudes and 32 fixed orientations, into held-out harmonic carriers. Exactly 200 CMB-only rows form the common reference set; 100 paired CMB-plus-noise rows and 799 CMB-only rows form two overlapping, non-independent evaluation arms. The positive-amplitude study contains 96,000 and 767,040 injected carriers, respectively, with no typed absences and no observation or raw-map access. The two arms give nearly the same cellwise power (median absolute difference 0.00271), but power is strongly template-, family-, reducer-, and combiner-dependent. In particular, the ECDF/min-local- p MES-10 functional is completely conservative on this 201-row design, whereas the exact integer-product combiner retains substantial power. These are powers of the registered 200-reference functionals, not power calibrations of the original 301- or 1000-row observation procedures and not sensitivities

to physical shear, vorticity, or Bianchi templates. Matched-null exchangeability applies only to the CMB-only evaluation arm; the paired CMB-plus-noise arm is a descriptive distribution-shift test against a CMB-only reference pool.

All fractions are conditional finite-pool ranks. They are not unconditional p -values, likelihoods, posterior probabilities, measurements of physical shear or vorticity, local/global attributions, foreground validations, or Bianchi-family identifications.

Independent-review verdict. The earlier component-replication review found $P0=0$ and $P1=0$. For the new carrier-intervention extension, one independent referee recomputed all 496 power rows without mismatch and found $P0=0$, two current-task P1 presentation and metadata defects, and two related P2 clarifications. This second revision uses the single allowed bounded repair to add the consumed claim-boundary metadata, state the paired-arm exchangeability failure locally, distinguish artifact-integrity replay from numerical regeneration, and replace the colliding heatmap with a readable 2×2 layout. Focused tests, exact content replay, and 3.3/6.8-inch visual inspection close both P1 findings; $P0=0$ and $P1=0$ remain.

The admitted evidence supports the stated one-way MES ceiling identities, conditional finite-pool scalar and observer-space morphology ranks, component- product robustness, external CMB holdouts, the fixed-bank foreground- projection diagnostic, and the registered 200-reference method-power result. None supports a physical inverse response, foreground-source attribution, family identification, or an unconditional detection claim. The scientific evidence cutoff is 29 August 2026; this typeset report is a synthesis of frozen results, not a redefinition of them. The method-power result is admitted only as a C2 conditional diagnostic for the registered 200-reference functional.

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1 Questions, scope, and logical layers

The analysis addresses three nested questions.

1. **MES premise layer.** Given registered almost-FRW and observer premises, what upper-ceiling coordinates follow from the measured low-order temperature multipole amplitudes?
2. **Observer-space layer.** What radial, shape, mixed, and parity-sensitive information is present in the measured $\ell = 2$ and $\ell = 3$ harmonic blocks themselves?
3. **Finite-pool layer.** How do the observed coordinates rank relative to predeclared, finite Planck simulation ensembles when the complete scoring operator is applied symmetrically to every row?

The arrows between these layers are deliberately one-way:

$$\text{measured sky} \longrightarrow \begin{cases} \text{scalar powers} \longrightarrow \text{MES ceilings,} \\ \text{harmonic blocks} \longrightarrow \text{observer-space STF tensors,} \\ \longrightarrow \text{finite-pool diagnostics.} \end{cases}$$

There is no arrow in this report from a scalar ceiling or an observer-space tensor to a physical spacetime state. Such an inverse would require a separately specified forward response, compatible frame and units, sufficient response rank, and additional observational support.

Table 1 separates the present theoretical and data-analysis results from work that has not been performed.

Table 1: Current result map. “Established” is always subject to the scope in the third column.

Object	Status	Exact scope or ceiling
MES ceiling identities	Established	Geodesic, almost-FRW, registered frame; residual cosmological $\epsilon_1 = 0$ for the principal reported branch
Scalar-to-vector/STF no-go	Established	Rotational scalars alone under an $O(3)$ -equivariant construction
Finite-rank validity	Conditional theorem	Joint exchangeability under the declared null plus row-permutation equivariance; conservative ties
Local/global identifiability	Structural result	Full nuisance-projected directional and depth-response rank is required
Scalar Planck analysis	Executed	One SMICA row plus exact paired 300-row CMB-plus-noise pool
Observer-space STF primary	Executed	Same observation plus paired 300-row pool; frame-free eight-coordinate family
CMB-only robustness lane	Executed	Same observation plus 999 CMB-only rows; separate diagnostic, not primary replacement
PR3 component replication	Executed	SMICA, Commander, NILC, and SEVEM; same paired null rows and frozen WU-006 statistic/operator
External CMB holdouts	Executed	WMAP9 ILC, CosmoGlobe CMB, and BeyondPlanck CMB under the same observation-row replacement contract
Foreground projection test	Executed with typed exclusions	Fixed bank of 13 Planck/CosmoGlobe/BeyondPlanck temperature templates; WMAP low-frequency, QUIJOTE, and CLASS exclusions are reported explicitly
Physical response inversion	Not performed	No shear, vorticity, local/global, or family inference

2 Cosmological and MES-bound framework

2.1 Kinematics and normalized target invariants

Let u^a be the registered fundamental-observer four-velocity, $u^a u_a = -1$, and

$$h_{ab} = g_{ab} + u_a u_b$$

the spatial projector. With metric signature $(-, +, +, +)$ and $\epsilon_{123} = +1$, the covariant velocity gradient is decomposed as

$$\nabla_b u_a = -u_b A_a + \frac{1}{3} \Theta h_{ab} + \sigma_{ab} + \omega_{ab},$$

where $A_a = u^b \nabla_b u_a$ is the acceleration, $\Theta = \nabla_a u^a$ is the expansion, $\sigma_{ab} = \sigma_{(ab)}$ is the spatial shear, and $\omega_{ab} = \omega_{[ab]}$ is the spatial vorticity. Define $H = \Theta/3$. The current MES coordinates target the squared dimensionless invariants

$$\Sigma^2 \equiv \frac{\sigma_{ab} \sigma^{ab}}{6H^2}, \quad W^2 \equiv \frac{\omega_{ab} \omega^{ab}}{6H^2}. \quad (1)$$

These are nonnegative scalar magnitudes. They are not vectors, principal axes, eigenframes, or signed tensors.

The MES programme uses the near-isotropy of the cosmic microwave background to place premise-dependent constraints on departures from an almost-FRW state [1, 2]. The present version uses the geodesic branch $A_a = 0$, the linear almost-EGS regime, and a registered fundamental-observer frame. Coefficients involving time or spatial derivatives of radiation multipoles inherit the corresponding derivative- hierarchy premises; they are not universal identities of arbitrary spacetimes.

2.2 Temperature multipole amplitudes

For a temperature anisotropy field with angular power C_ℓ , define

$$\epsilon_\ell = \frac{1}{T_0} \sqrt{\frac{(2\ell + 1)C_\ell}{4\pi}}, \quad (2)$$

where T_0 is expressed in the same temperature unit as $\sqrt{C_\ell}$. The reported numerical analysis fixes $T_0 = 2.72548 \text{ K} = 2.72548 \times 10^6 \mu\text{K}_{\text{CMB}}$. Thus every ϵ_ℓ is dimensionless and nonnegative. The observed dipole is dominated by observer motion. In the principal branch used below, that kinematic contribution is assigned to the observer-motion channel and the residual cosmological dipole is set to

$$\epsilon_1 = 0.$$

This is an explicit premise, not a conclusion extracted from the quadrupole or octupole.

2.3 Latest MES carriers and one-way ceilings

The registered geodesic carriers, before imposing $\epsilon_1 = 0$, are

$$B_\sigma = \frac{5}{3} \epsilon_1 + 3\epsilon_2 + \frac{3}{7} \epsilon_3, \quad (3)$$

$$B_\omega = \frac{10}{3} \epsilon_1 + \frac{2}{15} \epsilon_2. \quad (4)$$

On the principal branch they reduce to

$$B_\sigma = 3\epsilon_2 + \frac{3}{7}\epsilon_3, \quad B_\omega = \frac{2}{15}\epsilon_2.$$

The squared normalized ceiling coordinates are

$$\Sigma^2 \leq \Sigma_{\max}^2 = \frac{3}{2}B_\sigma^2 = \frac{27}{98}(7\epsilon_2 + \epsilon_3)^2, \quad (5)$$

$$W^2 \leq W_{\max}^2 = \frac{3}{2}B_\omega^2 = \frac{C_2}{30\pi T_0^2}. \quad (6)$$

The square symbols are essential: the reported variables are ceilings on the squared invariants in Eq. (1), not unsquared shear or vorticity amplitudes.

These inequalities are one-way. A sky realization produces ceiling coordinates after the premises have been declared. The converse—that a ceiling determines a unique physical source, geometry, direction, or spacetime—does not follow.

2.4 Invertibility and absence of information gain

On the registered nonnegative domain the principal-branch transformation can be inverted:

$$\epsilon_2 = \sqrt{\frac{75}{2}W_{\max}^2}, \quad (7)$$

$$\epsilon_3 = \sqrt{\frac{98}{27}\Sigma_{\max}^2 - 7\epsilon_2}, \quad (8)$$

$$C_\ell = \frac{4\pi T_0^2}{2\ell + 1}\epsilon_\ell^2. \quad (9)$$

Consequently, for any accompanying morphology vector M ,

$$\sigma(\Sigma_{\max}^2, W_{\max}^2, M) = \sigma(C_2, C_3, M) \quad (10)$$

on this domain. The MES coordinates can expose a useful physical normalization and can change the geometry of a chosen rank reducer, but they do not add statistically independent information to the same-row C_2, C_3 inputs.

2.5 Why a scalar ceiling cannot create direction or STF shape

Suppose s denotes any collection of rotation-invariant scalars and $F(s) \in \mathbb{R}^3$ is $O(3)$ -equivariant. Since $Rs = s$ for every rotation R ,

$$F(s) = F(Rs) = RF(s) \quad \text{for all } R \in O(3).$$

The only vector fixed by every rotation is zero. The same argument in the irreducible rank-two STF representation shows that an equivariant map from scalars alone to an STF tensor must also vanish. Therefore

$$\text{scalar MES ceiling} \not\rightarrow \text{direction, nonzero vector, or nonzero STF tensor.}$$

Direction-indexed data do permit legal moment maps. For a full-sky scalar field $q(\hat{n}, z)$,

$$V_a(z) = \frac{3}{4\pi} \int q(\hat{n}, z) n_a d\Omega, \quad (11)$$

$$T_{ab}(z) = \frac{15}{8\pi} \int q(\hat{n}, z) n_{\langle a} n_{b \rangle} d\Omega. \quad (12)$$

If $q = v_a n^a + S_{ab} n^a n^b$ plus harmonics orthogonal to $\ell = 1, 2$, with S_{ab} STF, these normalizations recover $V_a = v_a$ and $T_{ab} = S_{ab}$. For a masked or discretized sky, the corresponding estimator and quadrature must be bound explicitly. A single global scalar does not meet this premise.

2.6 Response-bound states and local/global identifiability

The current framework distinguishes three semantic objects:

Observable irrep state

Measured or simulated harmonic/STF content with a declared frame, basis, units, operator, and typed degenerate strata.

MES normalizer

A positive, same-channel, one-way ceiling that may rescale an already existing compatible amplitude. It creates no direction, shape, response, or information.

Response-bound state

An observable state tied to an independently specified forward response with compatible channel, frame, basis, units, epoch, and normalization. Its response rank determines point or set identification.

A rank-deficient response yields an identified set, not a point estimate. A simple local/global directional model illustrates the requirement:

$$V_i = f_L(z_i) \beta_L + f_G(z_i) \beta_G + \varepsilon_i. \quad (13)$$

Let S be the directional metric and K the 2×2 covariance-weighted depth Gram matrix for f_L, f_G after nuisance projection. The six-parameter Fisher matrix is $K \otimes S$, so

$$\det F = \det(K)^3 \det(S)^2. \quad (14)$$

One shell makes K rank one; proportional depth profiles do the same. Planck temperature morphology alone therefore cannot distinguish local boost from global tilt. No such physical inversion is reported below.

3 Data objects, common operator, and finite-pool statistics

3.1 Observation and two distinct null ensembles

The observation is the Planck PR3 SMICA temperature map. SMICA is a component-separated Planck product [3]; the large-angle statistical context and its sensitivity to masks and foreground treatment are described by the Planck isotropy analysis [4].

Two null ensembles have different scientific roles.

1. **Paired primary.** Exactly 300 ordered FFP10 SMICA CMB-plus-noise realizations, numbered 00000–00299, combined with the single observation to give $N = 301$ rows. This is the primary finite pool.
2. **CMB-only robustness lane.** Exactly 999 official FFP10 SMICA CMB-only realizations, numbered 00000–00999 with only 00970 absent and 00818 included, combined with the same observation to give $N = 1000$ rows. No noise realization enters this lane. It is a separate null-ensemble sensitivity check and does not replace the paired primary.

The common temperature operator performs one joint masked real-harmonic fit over $\ell = 0, \dots, 5$ and retains the 32 real coefficients at $\ell = 2, \dots, 5$. The registered frame is Galactic, the map ordering is HEALPix RING, and the unit is μK_{CMB} . The observed carrier was obtained in the same pass as its scalar features; it was neither reconstructed from those scalars nor estimated a

second time. Its scalar closure matches the frozen same-pass values, and the operator condition number is 6.2075, far below the predeclared ceiling 10^8 .

Exact row identity and deterministic replay establish what was compared; they do not establish that either simulation pool is a complete physical model of the observation.

3.2 Reducer A: leave-one-out absolute median

Let X_{ij} be coordinate j for row i . For a two-sided coordinate,

$$d_{ij} = |X_{ij} - \text{median}(X_{-i,j})|. \quad (15)$$

For an upper tail use $d_{ij} = X_{ij}$ and for a lower tail use $d_{ij} = -X_{ij}$. The observation-inclusive local rank is

$$p_{ij} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(d_{kj} \geq d_{ij}). \quad (16)$$

Because every row has its own leave-one-out centre, the complete map from all rows to all scores, not only the observation-row formula, is the statistical object. This reducer is invariant under positive affine rescaling but not under arbitrary monotone reparameterization.

3.3 Reducer B: leave-one-out ECDF midrank

For every row and coordinate define

$$u_{ij} = \frac{\#\{k \neq i : X_{kj} < X_{ij}\} + \frac{1}{2}\#\{k \neq i : X_{kj} = X_{ij}\}}{N-1}. \quad (17)$$

The oriented score is $2|u_{ij} - 1/2|$ for a two-sided coordinate, u_{ij} for an upper tail, and $1 - u_{ij}$ for a lower tail. Exact integer midrank numerators preserve mathematical ties. This reducer is invariant under strictly increasing coordinate transformations that preserve tie structure.

3.4 Family scan, ties, and the finite-rank premise

For a predeclared family $\mathcal{F}$, every row receives

$$r_i = \min_{j \in \mathcal{F}} p_{ij}, \quad S_i = -\log r_i. \quad (18)$$

The reported family rank is

$$p_{\mathcal{F}} = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(r_k \leq r_0) = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{1}(S_k \geq S_0). \quad (19)$$

Ties and the observation itself are included, so a zero rank is impossible; the resolution floors are $1/301$ and $1/1000$ [5].

If $(Z_0, \dots, Z_{N-1})$ are jointly exchangeable under the stated null and the complete row-scoring map is permutation-equivariant, then the observation-inclusive family rank is super-uniform, with ties treated conservatively. Row-swap and row-permutation checks establish the second condition. The first condition is a scientific null-model premise. If FFP10 does not faithfully represent the observation under the intended null, the fractions remain exact ranks relative to the frozen pool but do not acquire unconditional Type-I-error calibration.

The proof is short. Exchangeability of the input rows and equivariance of the complete map imply exchangeability of $(S_0, \dots, S_{N-1})$. Conditional on their unordered multiset, the observation label

is uniformly distributed over the N positions. Counting all scores at least as large as S_0 , including ties, can only increase the ordinary randomized rank; hence the reported rank is super-uniform.

Feature families, coordinate transformations, tails, reducers, missing-row rules, and the ϵ_1 grid were fixed before observation-row scoring. No covariance matrix, Gaussian likelihood, posterior, Bayes factor, or observation-selected feature subset enters Eq. (19).

4 Scalar MES and morphology results

4.1 The eight non-anchor morphology coordinates

The scalar analysis combines two low-order amplitude or MES coordinates with eight morphology coordinates. Define the even-to-odd power ratio

$$\mathcal{P} = \frac{C_2 + C_4}{C_3 + C_5}. \quad (20)$$

For each ℓ , form the real symmetric angular-momentum power tensor

$$\tilde{A}_{ij}^{(\ell)} = \frac{1}{2\ell(\ell+1)} \sum_{m,m'} a_{\ell m}^* \{J_i, J_j\}_{mm'} a_{\ell m'}, \quad A^{(\ell)} = \frac{\tilde{A}^{(\ell)}}{\text{tr } \tilde{A}^{(\ell)}}. \quad (21)$$

If $\lambda_{\ell,1} \leq \lambda_{\ell,2} \leq \lambda_{\ell,3}$ are its trace-one eigenvalues, the registered gaps are

$$G_\ell = \lambda_{\ell,3} - \lambda_{\ell,2}, \quad \ell = 2, 3.$$

Multipole vectors are the headless axes in the harmonic-polynomial factorization

$$T_\ell(\hat{n}) = A_\ell \left[\prod_{r=1}^{\ell} (\mathbf{v}_{\ell,r} \cdot \hat{n}) - \text{traces} \right], \quad (22)$$

defined up to permutation and individual sign changes [6, 7]. Sign-invariant coordinates are

$$d_2 = |\mathbf{v}_{2,1} \cdot \mathbf{v}_{2,2}|, \\ (d_{3,0}, d_{3,1}, d_{3,2}) = \text{sort}(|\mathbf{v}_{3,1} \cdot \mathbf{v}_{3,2}|, |\mathbf{v}_{3,1} \cdot \mathbf{v}_{3,3}|, |\mathbf{v}_{3,2} \cdot \mathbf{v}_{3,3}|).$$

Let $\hat{\mathbf{n}}_2$ be the normalized quadrupole plane normal and $\hat{\mathbf{n}}_{3,ij}$ the three normalized octupole pair normals. The plane-alignment coordinate is

$$A_{23} = \max_{i < j} |\hat{\mathbf{n}}_2 \cdot \hat{\mathbf{n}}_{3,ij}|. \quad (23)$$

The eight-coordinate morphology block is therefore $M = (\mathcal{P}, G_2, G_3, d_2, d_{3,0}, d_{3,1}, d_{3,2}, A_{23})$.

4.2 Six paired family decompositions

Six families were compared on the identical 301 rows with the original all-two-sided, leave-one-out absolute-median operator:

Family	First coordinates	Remaining coordinates
Generic 12	C_2, C_3, C_4, C_5	M
Raw reduced 10	C_2, C_3	M
Linear- ϵ 10	ϵ_2, ϵ_3	M
MES-squared 10	$\Sigma_{\max}^2, W_{\max}^2$	M
Anchors only 2	$\Sigma_{\max}^2, W_{\max}^2$	none
Morphology only 8	none	M

Table 2: Exact scalar family ranks in the paired 300-row primary pool. The families are a paired operator decomposition, not six independent tests.

Family	Global rank	Observation-row minimum
Generic 12	133/301	$d_{3,0} : 16/301$
Raw reduced 10	110/301	$d_{3,0} : 16/301$
Linear- ϵ 10	109/301	$d_{3,0} : 16/301$
MES-squared 10	98/301	$d_{3,0} : 16/301$
Anchors only 2	78/301	both anchors: 74/301
Morphology only 8	88/301	$d_{3,0} : 16/301$

The observed squared ceilings are

$$\Sigma_{\max}^2 = 2.4000870222 \times 10^{-10}, \quad W_{\max}^2 = 3.0061887931 \times 10^{-13}. \quad (24)$$

They follow from

$$C_2 = 210.4618472 \mu\text{K}_{\text{CMB}}^2, \quad C_3 = 482.0208114 \mu\text{K}_{\text{CMB}}^2,$$

which give $\epsilon_2 = 3.3575598243 \times 10^{-6}$ and $\epsilon_3 = 6.0122078079 \times 10^{-6}$ through Eq. (2). Both have local rank 74/301. Their Pearson correlation over the complete pool is 0.994874, their Spearman correlation is 0.993738, and the two-coordinate correlation block has condition number 389.2. These facts quantify strong same-row dependence; they are not evidence for two independent measurements.

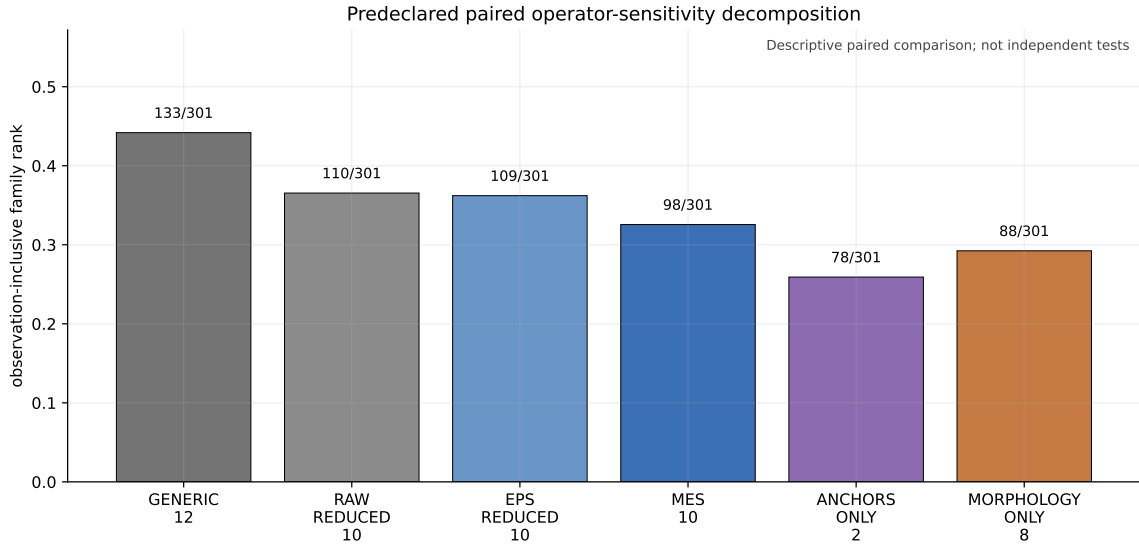


Figure 1: Direct frozen analysis output for the six scalar families. The largest step from the generic control occurs when C_4, C_5 are removed. The observation-row minimum remains the octupole morphology coordinate $d_{3,0}$, not either MES ceiling. Differences among bars are descriptive paired operator sensitivity.

Thus the statement supported by the data is narrow: the observation-row minimum is morphology-driven rather than anchor-driven. The complete family rank nevertheless depends on the coordinate family because adding or replacing coordinates changes the row-score distribution for every null row.

4.3 Factorial coordinate-mechanism audit

To separate nonlinear squaring from carrier mixing, four ten-coordinate cells were fixed on the same 301 rows:

$$\begin{aligned} \text{linear } \epsilon &: (\epsilon_2, \epsilon_3, M), \\ \text{square only} &: (\epsilon_2^2, \epsilon_3^2, M), \\ \text{carrier only} &: (B_\sigma, B_\omega, M), \\ \text{MES squared} &: (\tfrac{3}{2}B_\sigma^2, \tfrac{3}{2}B_\omega^2, M). \end{aligned}$$

Table 3: Factorial family ranks under the two registered reducers and the all-two-sided tail registry.

Cell	Absolute-median rank	ECDF-midrank rank
Linear ϵ	109/301	153/301
Square only	110/301	153/301
Carrier only	98/301	148/301
MES squared	98/301	148/301

For the absolute-median reducer, the carrier-mixing contrasts are -11 and -12 rank counts in the linear and squared branches, while the squaring contrasts are $+1$ without mixing and 0 with mixing; the interaction is -1 . For the ECDF reducer, mixing changes the rank by -5 in both branches, while squaring and the interaction are zero. The main lesson is not that one cell is a new test. It is that the legacy $98/301$ value is partly a property of the selected coordinate mechanism and reducer.

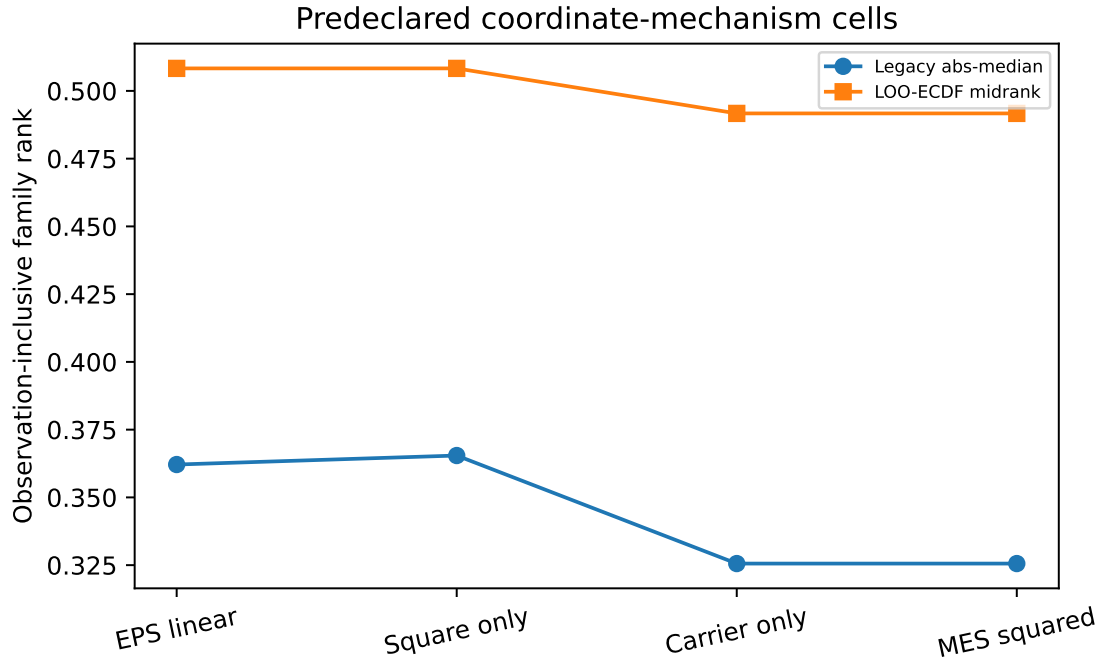


Figure 2: Direct frozen factorial result. Each line compares the same four coordinate cells under one reducer. The separation between reducer lines is a method-sensitivity result, not a change in the observed sky.

4.4 Tail-registry sensitivity

The factorial comparison above retains the all-two-sided tail convention so that coordinate-mechanism effects are isolated. A separate, predeclared physics-oriented registry uses upper tails for the two MES ceilings, the two power-tensor gaps, and the plane-alignment maximum, while retaining two-sided tails for parity and all four multipole-vector dot products. Applied to the MES-squared cell, the results are

Tail registry	Absolute median	ECDF midrank
All two-sided	98/301	148/301
Physics-oriented	98/301	93/301

The unchanged absolute-median rank and changed ECDF rank are another paired operator-sensitivity result. Neither row is an independently selected test, and the 93/301 value does not supersede the all-two-sided factorial result.

4.5 Residual-dipole premise sensitivity

Equations (3)–(4) were evaluated rowwise on a predeclared 101-point grid $0 \leq \epsilon_1 \leq 10^{-5}$. The observed ceilings change substantially over that interval: at the upper endpoint, $\Sigma_{\max}^2$ is 5.37 times and $W_{\max}^2$ is 5694 times their $\epsilon_1 = 0$ values. Nevertheless the complete MES-squared family ranks remain 98/301 and 148/301 throughout the grid for the two reducers.

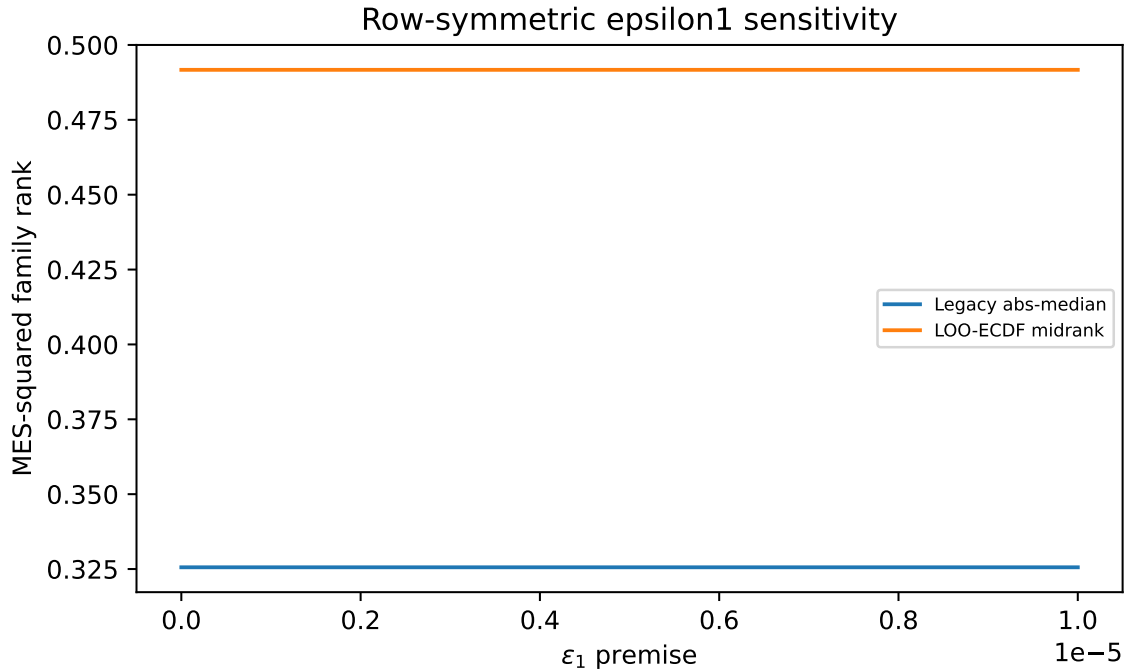


Figure 3: Predeclared residual-dipole scan. Absolute ceiling magnitudes vary, whereas the two complete-pool family ranks are constant on this grid. This is a property of this frozen family, rowwise transformation, grid, and two reducers; it is not a general theorem of dipole insensitivity.

5 Observer-space irreducible tensors

5.1 Harmonic convention and reversible STF bridge

For a real temperature map the retained convention is

$$T_\ell(\hat{n}) = a_{\ell 0} Y_{\ell 0}(\hat{n}) + 2 \sum_{m=1}^{\ell} \Re[a_{\ell m} Y_{\ell m}(\hat{n})], \quad (25)$$

with orthonormal Condon–Shortley harmonics. The real storage block is

$$(a_{\ell 0}, \Re a_{\ell 1}, \Im a_{\ell 1}, \dots, \Re a_{\ell \ell}, \Im a_{\ell \ell}).$$

The $\ell = 2$ block is mapped linearly to a real symmetric trace-free tensor Q_{ab} , and the $\ell = 3$ block to a real fully symmetric trace-free tensor O_{abc} [8]. Five and seven independent Cartesian components respectively match the dimensions $2\ell + 1$. The explicit normalization is given in Appendix A. Every row passes

$$a_{\ell m} \longrightarrow (Q, O) \longrightarrow a_{\ell m}$$

with maximum absolute residual $1.4210854715 \times 10^{-14}$ against the predeclared $\text{atol} = \text{rtol} = 5 \times 10^{-13}$. This validates the numerical representation bridge, not a physical source model.

5.2 Radial, shape, mixed, and parity coordinates

The radial and quadrupole-shape invariants are

$$q_2 = Q_{ab} Q^{ab}, \quad q_3 = Q_a{}^b Q_b{}^c Q_c{}^a, \quad (26)$$

$$o_2 = O_{abc} O^{abc}, \quad J_Q = \sqrt{6} \frac{q_3}{q_2^{3/2}}. \quad (27)$$

Here q_2, o_2 have units $\mu\text{K}_{\text{CMB}}^2$, q_3 has units $\mu\text{K}_{\text{CMB}}^3$, and J_Q is dimensionless. For a real symmetric trace-free 3×3 tensor,

$$Q^3 - \frac{q_2}{2} Q - \frac{q_3}{3} I = 0, \quad \frac{q_2^3}{2} - 3q_3^2 \geq 0,$$

so $J_Q \in [-1, 1]$ when $q_2 > 0$.

When $q_2, o_2 > 0$, define

$$\hat{Q} = Q/\sqrt{q_2}, \quad \hat{O} = O/\sqrt{o_2}, \quad v_a = \hat{O}_{abc} \hat{Q}^{bc},$$

and

$$S_{ab} = \text{STF}(\hat{O}_{acd} \hat{O}_b{}^{cd}).$$

The registered dimensionless mixed coordinates are

$$\hat{R}_{v0} = v_a v^a, \quad \hat{R}_{v1} = v_a \hat{Q}^a{}_b v^b, \quad (28)$$

$$\hat{R}_{v2} = v_a (\hat{Q}^2)^a{}_b v^b, \quad \hat{R}_{QS} = \hat{Q}_{ab} S^{ab}, \quad (29)$$

$$\hat{K}_v = \det[v, \hat{Q}v, \hat{Q}^2 v]. \quad (30)$$

The first four are reflection-even; $\hat{K}_v$ is a pseudoscalar that changes sign under reflection. It is therefore an $SO(3)$ invariant with explicit parity covariance, not an ordinary $O(3)$ scalar.

The frame-free family is

$$\mathcal{F}_{\text{ff}} = \{q_2, o_2, J_Q, \hat{R}_{v0}, \hat{R}_{v1}, \hat{R}_{v2}, \hat{R}_{QS}, \hat{K}_v\}. \quad (31)$$

For a separate absolute-frame sensitivity check, let $z_G = (0, 0, 1)$ be the Galactic pole and define

$$P_G^G = |\hat{Q} z_G|^2, \quad P_O^G = |\hat{O}_{abc} z_G^b z_G^c|^2.$$

The ten-coordinate companion is $\mathcal{F}_G = \mathcal{F}_{\text{ff}} \cup \{P_Q^G, P_O^G\}$. It is descriptive and is not a physical source-direction estimator.

5.3 Degenerate strata and typed absence

At $q_2 = 0$ the quadrupole shape is undefined; at $o_2 = 0$ the octupole shape and mixed coordinates are undefined. Repeated quadrupole eigenvalues and rank-deficient Krylov matrices can also make a registered orientation or partial-domain coordinate unavailable. Such cases are represented as typed absence rather than by an invented fallback axis or silent zero filling. A missing primary-family value would stop the family calibration. All 301 rows of the paired pool and all 1000 rows of the CMB-only analysis are complete on the registered primary family; every quadrupole spectrum used here is simple. Global completeness of this coordinate catalogue over all tensor orbits has not been proven.

6 Paired 300-row observer-space result

6.1 Observed coordinates and local ranks

Table 4: Observed frame-free coordinates and local ranks in the paired primary. The ranks use denominator 301. q_3 is algebraic audit information and is not scored separately.

Coordinate	Observed value	Abs.-median	ECDF-midrank
q_2	1187.746699	86/301	34/301
q_3	−1237.223041	—	—
J_Q	−0.0740351	248/301	248/301
o_2	9340.678306	241/301	242/301
$\hat{R}_{v0}$	0.4727877	4/301	4/301
$\hat{R}_{v1}$	−0.0715848	114/301	118/301
$\hat{R}_{v2}$	0.2371130	4/301	4/301
$\hat{R}_{QS}$	9.9376×10^{-4}	282/301	284/301
$\hat{K}_v$	−0.0053223	80/301	80/301

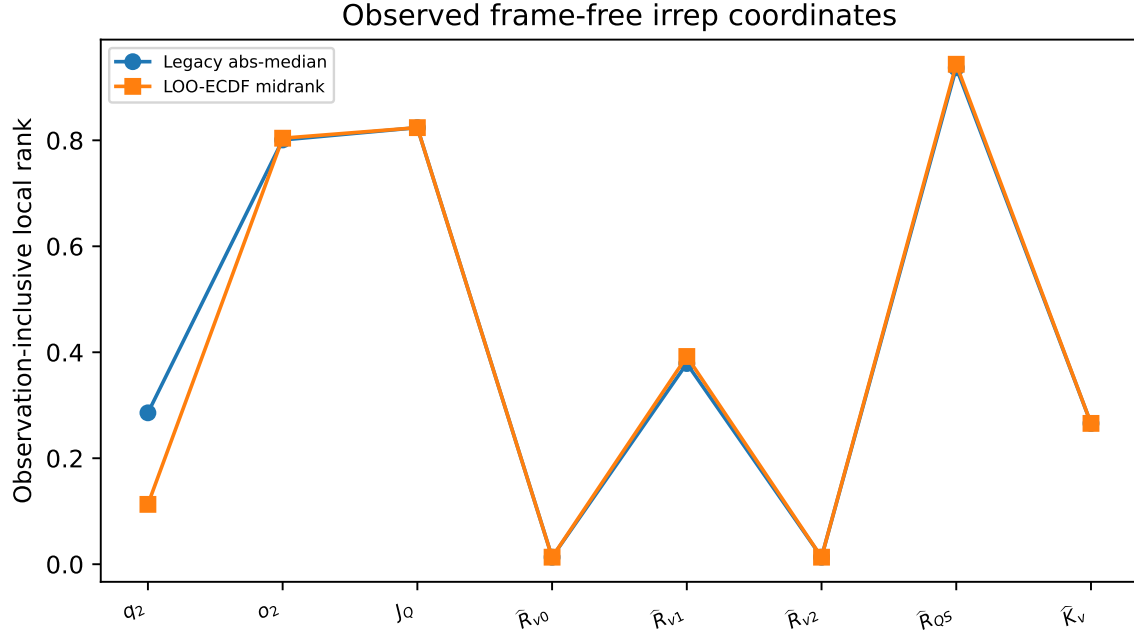


Figure 4: Direct frozen paired-pool output. The smallest observation-row local ranks are 4/301 for the unsigned mixed concentrations $\hat{R}_{v0}$ and $\hat{R}_{v2}$. The different q_2 ranks illustrate reducer sensitivity. Local minima are not family-level detections.

6.2 Family ranks and dependence

Table 5: Observer-space family ranks in the paired primary.

Family	Absolute median	ECDF midrank
Frame-free primary, 8D	27/301	27/301
Galactic companion, 10D	34/301	33/301

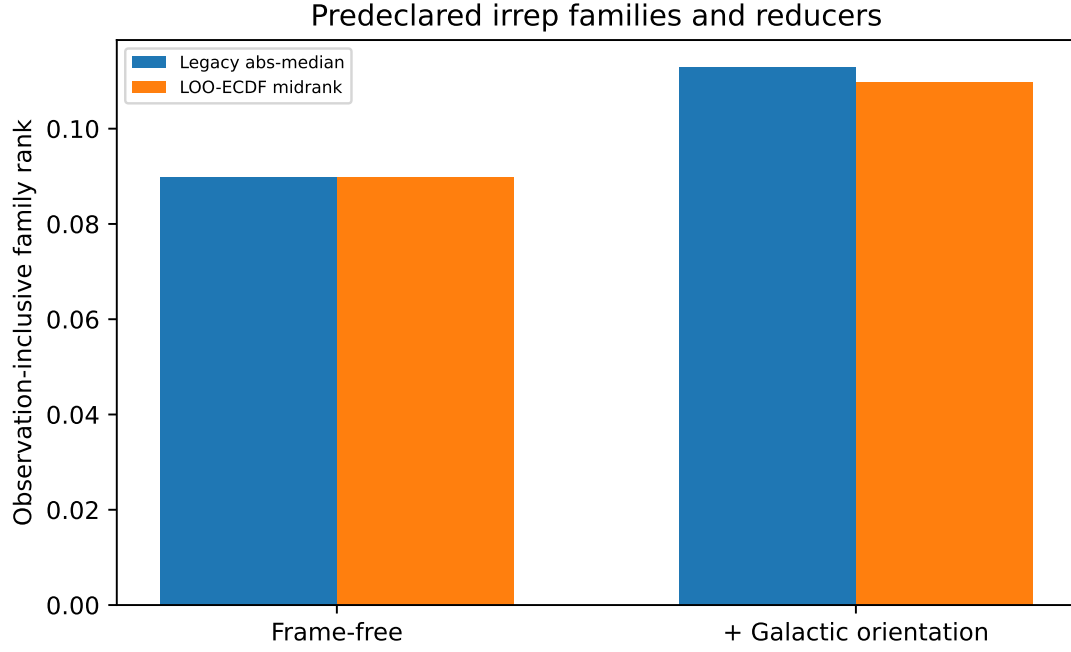


Figure 5: Direct frozen paired-pool family comparison. Adding two Galactic-pole coordinates changes the full row-score distribution. The difference is an absolute-frame sensitivity, not evidence for a Galactic physical source.

The observation-row minimum is $4/301$, giving $S_0 = -\log(4/301) = 4.3208159$, but 27 complete rows are at least as extreme after every row is allowed to select its own minimum coordinate. The family rank is therefore $27/301$, not $4/301$.

The Spearman-correlation participation ratios are 6.5815 for the eight frame-free coordinates and 8.4260 for the ten-coordinate companion. They summarize dependence spread and are not numbers of independent experiments or physical degrees of freedom.

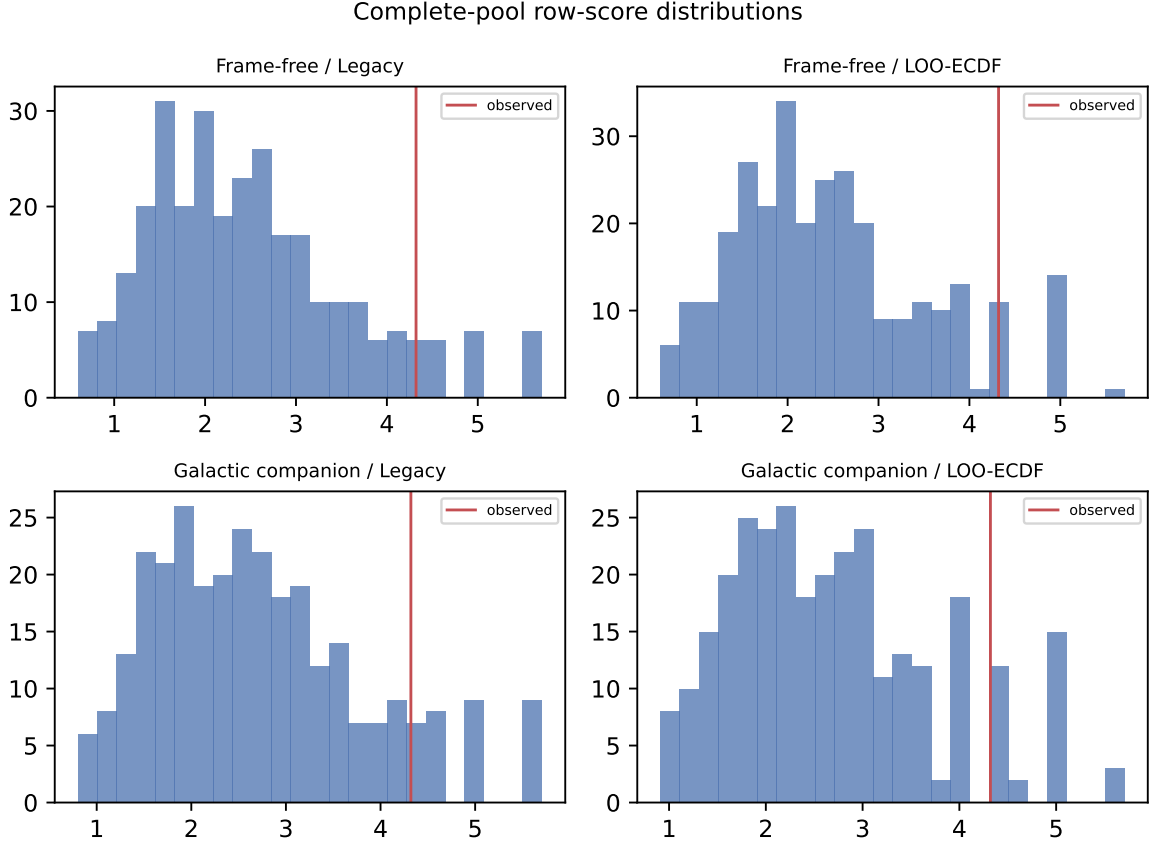


Figure 6: The 300 paired-null maximum-score distributions with the observed score marked vertically. Histograms are visualizations of the same exact discrete comparison in Table 5; they are not fitted densities or additional significance calculations.

7 Frozen component replication and foreground-projection test

7.1 Execution contract and interpretation boundary

This extension changes exactly one object at a time: the observed temperature map supplying row zero. It preserves the WU-006 joint-cut-sky mask and operator, harmonic basis, $\ell = 2, \dots, 5$ transfer, the exact ordered 300 paired CMB-plus-noise carrier rows, both reducers, all tails, both family memberships, and the observation-inclusive row-scoring rule. Thus every admitted product is analyzed as

$$(Z_0^{(p)}, Z_1, \dots, Z_{300}),$$

where only $Z_0^{(p)}$ depends on product p . No null map was reopened and the accepted SMICA carrier was reproduced numerically before replacement. Delivered PR3 component-product beams were recorded but not applied, because applying a product-specific transfer after inspecting the products would have changed the frozen operator.

This design isolates sensitivity to the observation product, but it does not make the Planck FFP10 pool exchangeable with WMAP, CosmoGlobe, or BeyondPlanck. Consequently the Planck component ranks are conditional replications against the original frozen pool, while the external-product ranks are descriptive holdout scores against that pool, not calibrated cross-instrument p -values.

7.2 PR3 component products and external CMB holdouts

Table 6 reports the radial invariants q_2 and o_2 , the two coordinates carrying the smallest SMICA local ranks, $\hat{R}_{v0}$ and $\hat{R}_{v2}$, and the frame-free and Galactic family ranks. Each rank pair is absolute-median/ECDF-midrank. The full Q_{ab} and O_{abc} component vectors are retained in the generated cross-product table; q_2 and o_2 summarize their squared norms in the registered orthonormal real-harmonic basis.

Table 6: Cross-product stability under exact observation-row replacement. All ranks have denominator 301. “FF” is the eight-coordinate frame-free family and “G” is the ten-coordinate Galactic companion.

Product	q_2	o_2	$\hat{R}_{v0}$	$\hat{R}_{v2}$	$\cos^2_Q$	$\cos^2_O$	FF L/E	G L/E
SMICA	1187.747	9340.678	0.472788	0.237113	1.000000	1.000000	27/27	34/33
Commander	1040.589	8815.922	0.442541	0.225185	0.988573	0.996828	46/41	58/53
NILC	1208.299	9502.155	0.479976	0.237819	0.993741	0.999799	27/27	34/33
SEVEM	756.295	9069.171	0.413061	0.167852	0.821135	0.913665	81/40	102/50
WMAP9 ILC	1425.999	10272.120	0.471676	0.248155	0.969095	0.994801	15/15	19/18
CosmoGlobe CMB	12037.663	20587.512	0.355955	0.185962	0.468890	0.573292	69/89	89/112
BeyondPlanck CMB	5543.807	14158.920	0.364681	0.199840	0.594981	0.714986	64/62	83/78

Here $\cos^2_Q$ and $\cos^2_O$ are squared Frobenius cosines with the SMICA STF tensors. They compare shape while discarding overall amplitude and sign. Commander and NILC closely track both SMICA tensor shapes; SEVEM is less close, especially in Q . The family result is stable for NILC, moderately shifted for Commander, and reducer-sensitive for SEVEM. None of those descriptive differences is a test of the difference between products.

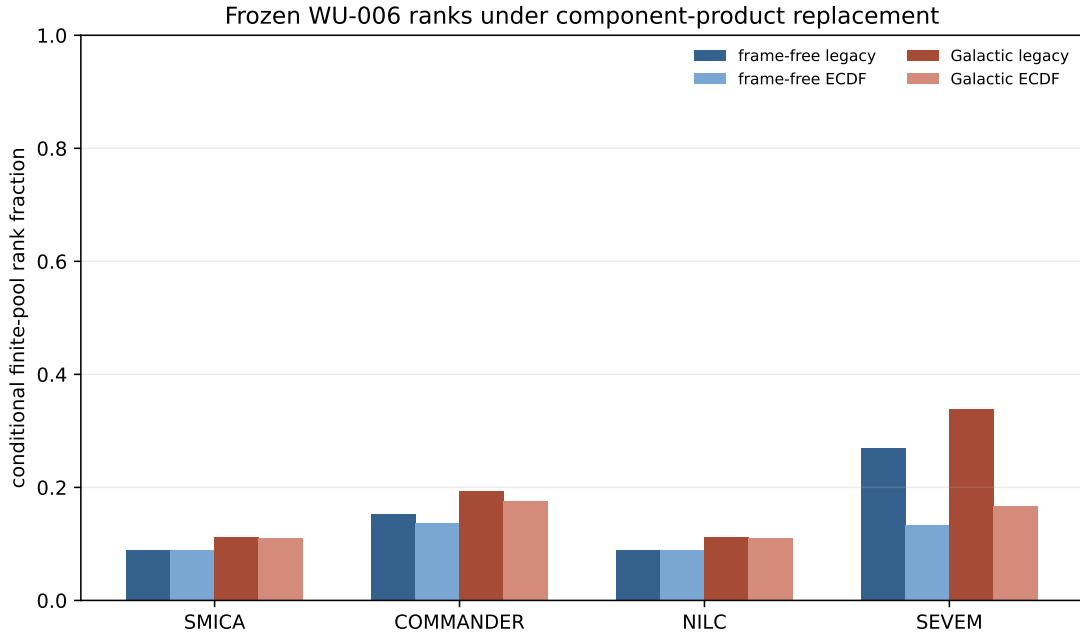


Figure 7: Frozen family ranks for the four PR3 component-separated products. The same 300 null rows score every replacement observation. Bar differences are sensitivity diagnostics, not independent significances.

Table 7 gives every frame-free local rank. A cell a/b means absolute-median numerator a and ECDF-midrank numerator b , always over 301 rows. This exposes coordinate-level changes that a single family fraction would hide.

Table 7: All eight coordinate-rank numerators under component replacement and external CMB holdout. Column labels follow Eq. (31).

Product	q_2	o_2	J_Q	$\hat{R}_{v0}$	$\hat{R}_{v1}$	$\hat{R}_{v2}$	$\hat{R}_{QS}$	$\hat{K}_v$
SMICA	86/34	241/242	248/248	4/4	114/118	4/4	282/284	80/80
Commander	73/14	267/274	194/194	10/10	76/78	7/7	290/286	63/64
NILC	87/36	234/236	275/274	4/4	206/194	4/4	227/224	66/66
SEVEM	65/6	253/254	162/162	14/14	24/30	27/27	94/96	42/40
WMAP9 ILC	101/54	200/202	131/134	4/4	33/36	2/2	217/216	72/76
CosmoGlobe CMB	23/46	18/36	219/222	33/33	12/16	17/17	217/216	193/184
BeyondPlanck CMB	227/234	64/102	140/140	28/28	12/16	11/11	132/124	62/64

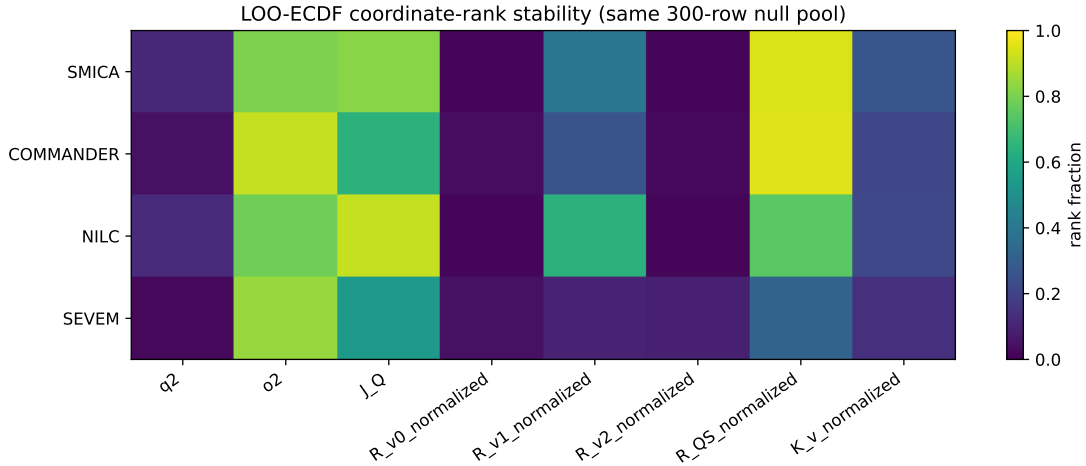


Figure 8: LOO-ECDF coordinate-rank fractions for the four PR3 products. The two small SMICA mixed-coordinate ranks persist in NILC and remain low, though not identical, in Commander and SEVEM.

The WMAP9 ILC holdout is especially close in tensor shape and has frame-free ranks 15/301 under both reducers. This is useful transport evidence but not an independent calibration, because the null rows remain Planck FFP10. The CosmoGlobe and BeyondPlanck CMB products differ more strongly in amplitudes and shape and have less extreme family ranks. The variation demonstrates why the SMICA result cannot be promoted without product and null-fidelity qualifiers.

7.3 Fixed-bank foreground falsification

To test a simple projection explanation without fitting after seeing SMICA, the foreground template bank was fixed in advance. For every SMICA or paired null row i and template t , define

$$A_{it}^Q = \frac{\langle Q_i, Q_t \rangle_F^2}{\|Q_i\|_F^2 \|Q_t\|_F^2}, \quad A_{it}^O = \frac{\langle O_i, O_t \rangle_F^2}{\|O_i\|_F^2 \|O_t\|_F^2},$$

and the simultaneous single-template score

$$S_i^{\text{fg}} = \max_{t \in \mathcal{T}} \min(A_{it}^Q, A_{it}^O).$$

The upper-tail rank counts every row whose own maximum over the same fixed bank is at least the observed maximum. Thus the look-elsewhere choice of template is included inside the row-symmetric score.

The admitted bank contains Planck PR4 30, 44, 353, 545, and 857 GHz maps plus synchrotron, dust, AME, and free-free component maps from each of CosmoGlobe and BeyondPlanck: 13 templates

in total. The best SMICA template is the CosmoGlobe free-free map, but its simultaneous score is only 0.072797. The fixed-bank rank is

$$144/301.$$

The predeclared exceptional-alignment gate was $\text{numerator} \leq 15$. Therefore the necessary morphology expected from a simple single-template projection is not exceptional in this bank. This result actively disfavors that narrow explanation; it does not exclude mixtures, nonlinear leakage, transfer mismatch, an omitted foreground morphology, or other systematics, and it does not validate a physical source.

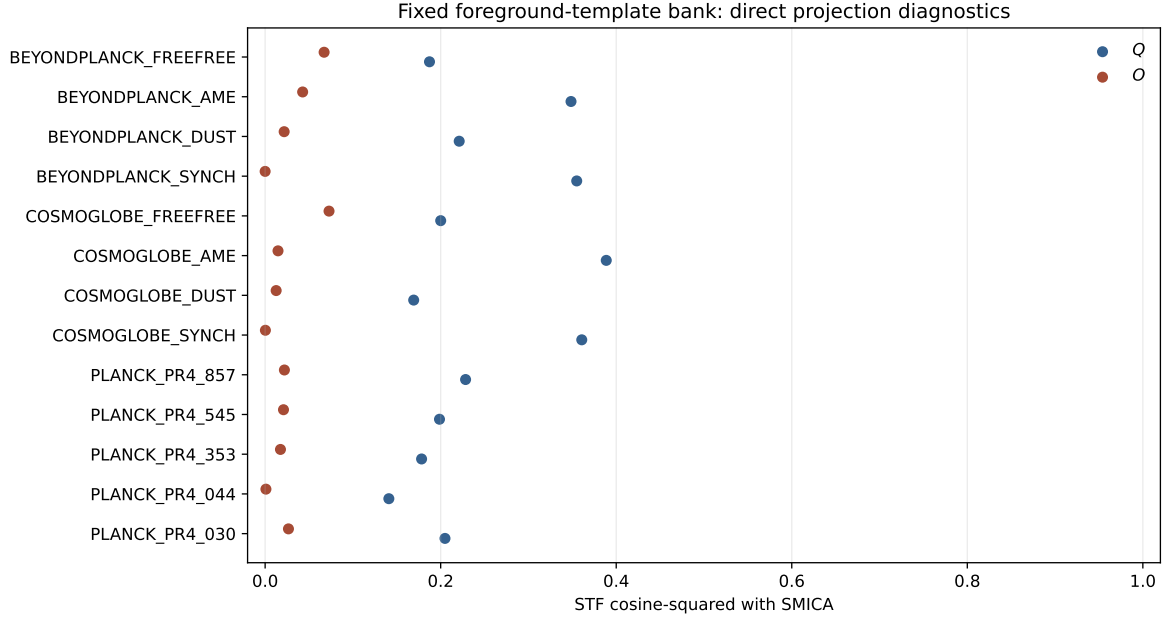


Figure 9: Squared STF cosines between SMICA and each admitted foreground template. A single-template projection would require simultaneous Q and O alignment; the bank statistic uses the smaller coordinate and calibrates the maximum over templates against every paired-null row.

The remaining requested lanes were not silently coerced into the bank. WMAP9 K/Ka/Q temperature maps were opened read-only but refused because their FITS unit string is `mK,thermodynamic`, whereas the predeclared registry expected `mK, thermodynamic`; this is a representation-level admission mismatch, not evidence of scientific disagreement, and no WMAP low-frequency value was scored. QUIJOTE 11/13/17/19 GHz products contain partial-sky UNSEEN support incompatible with the frozen Planck mask; filling or changing the mask was forbidden. The available CLASS 40 GHz and reobserved products provide V, Q, U rather than a compatible temperature- I channel, so they cannot generate the temperature Q_{ab}, O_{abc} objects without fabrication. No compatible component split was locally available. Each is a typed lane limitation, not a failure of the successful products.

8 Separate 999-row CMB-only robustness result

The same observed carrier, feature registry, tails, and reducers were compared with the distinct CMB-only ensemble. The observation map was not re-estimated; 999 CMB-only null maps were processed, with no noise maps read. The paired 300-row result remains the primary result.

Table 8: Observer-space family ranks in the 999-row CMB-only robustness pool.

Family	Absolute median	ECDF midrank
Frame-free primary, 8D	61/1000	55/1000
Galactic companion, 10D	78/1000	73/1000

The frame-free local-rank numerators, in the coordinate order of Eq. (31), are

$$\begin{aligned} (245, 859, 911, 9, 370, 21, 992, 294) & \quad (\text{absolute median}), \\ (90, 870, 916, 9, 356, 21, 992, 300) & \quad (\text{ECDF midrank}). \end{aligned}$$

The minimum is 9/1000 at $\hat{R}_{v0}$, followed by 21/1000 at $\hat{R}_{v2}$. Complete-family calibration raises these to the ranks in Table 8.

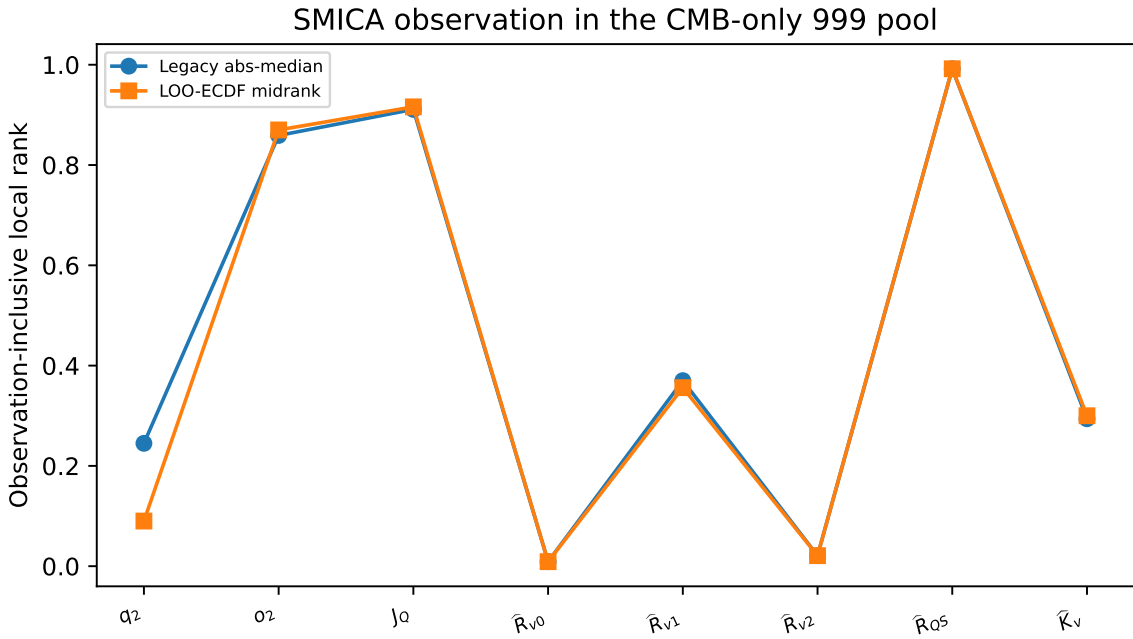


Figure 10: Direct frozen CMB-only robustness output. The observation is fixed; only the finite comparison ensemble differs from the paired primary.

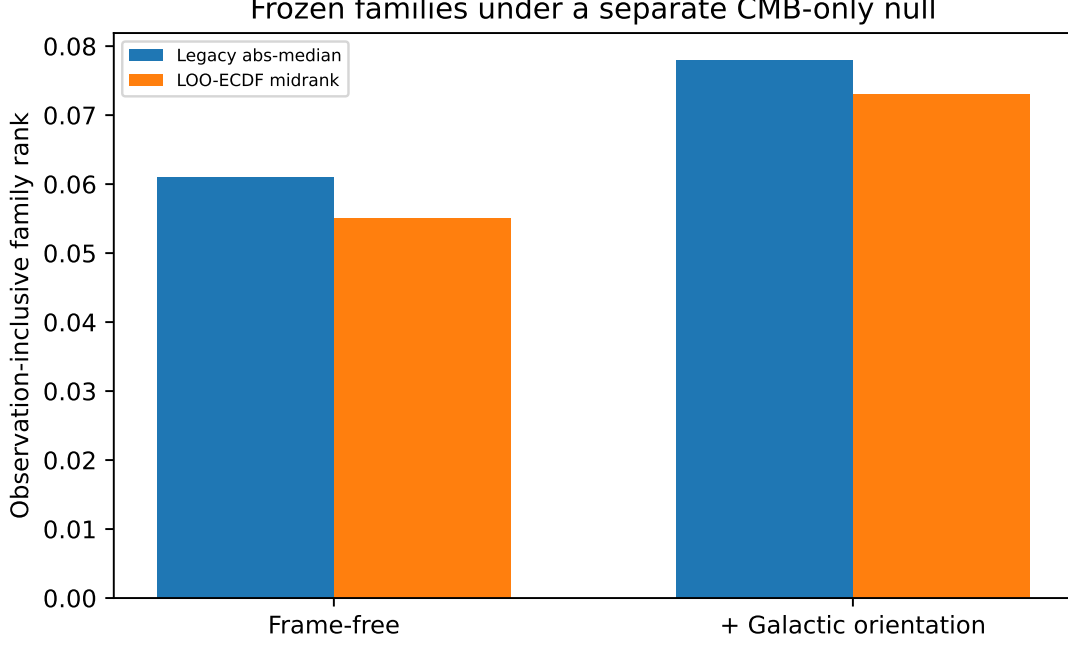


Figure 11: CMB-only family ranks. Lower fractions than in the paired primary show null-ensemble sensitivity. They do not validate foreground cleaning, constitute an independent test, or promote a detection claim.

For the frame-free family, the CMB-only minus paired rank-fraction differences are -0.02870 and -0.03470 under the two reducers. For the Galactic companion they are -0.03496 and -0.03663 . These comparisons reuse the same observation and feature families, so they are descriptive paired sensitivities, not independent replications. The CMB-only pool omits instrument noise by construction and cannot establish foreground or component-separation fidelity.

The CMB-only participation ratios are 6.6625 for the frame-free family and 8.5474 for the Galactic companion. The STF round-trip residual remains $1.4210854715 \times 10^{-14}$ under the same frozen criterion.

9 Registered carrier-domain method-power experiment

9.1 Question and fixed intervention

The preceding sections ask how the observed sky ranks in frozen finite pools. The present experiment asks a different question: if a predeclared algebraic pattern is added to otherwise held-out simulation carriers, how often does a specified 200-reference scoring functional reject at level 10/201? The observation is never used. This is a method-power experiment, not a new observation test and not an estimate of physical-source sensitivity.

Write each real harmonic block in the registered layout

$$\mathbf{a}_\ell = (a_{\ell 0}, \Re a_{\ell 1}, \Im a_{\ell 1}, \dots, \Re a_{\ell \ell}, \Im a_{\ell \ell})$$

with metric $G_\ell = \text{diag}(1, 2, 2, \dots, 2)$. The full carrier has 32 components for $\ell = 2, \dots, 5$. Five integer-seeded templates are registered: Q_AXIAL, O_PLANAR, MIX_AXIAL, MIX_PLANAR, and MIX_GENERIC. Each is normalized so that

$$\mathbf{t}_2^\top G_2 \mathbf{t}_2 + \mathbf{t}_3^\top G_3 \mathbf{t}_3 = 1, \quad \mathbf{t}_4 = \mathbf{t}_5 = 0,$$

with equal energy in the two active blocks when both are nonzero. They are analytic carrier perturbations. They are expressly not shear, vorticity, or Bianchi solutions.

Thirty-two deterministic proper rotations are obtained from hash-fixed open unit-cube points and unit quaternions. If $D_\ell(R)$ is the metric-preserving real-harmonic representation, the intervention is

$$\mathbf{a}'_{2,3} = \mathbf{a}_{2,3} + A s \sqrt{4\pi} D(R) \mathbf{t}_{2,3}, \quad \mathbf{a}'_{4,5} = \mathbf{a}_{4,5}, \quad (32)$$

at $A \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$. The shared scale is fixed from CMB-only reference rows 00000–00199:

$$s = \text{median}_i \left[\frac{\mathbf{a}_{2,i}^\top G_2 \mathbf{a}_{2,i} + \mathbf{a}_{3,i}^\top G_3 \mathbf{a}_{3,i}}{4\pi} \right]^{1/2} = 34.0422165082 \mu\text{K}_{\text{CMB}}.$$

Zero amplitude is computed once per arm; repeating it over templates or orientations is forbidden because that would manufacture trials.

9.2 Reference, evaluation arms, and exact finite scores

The reference set is the same 200 CMB-only carriers in both arms. The paired arm evaluates IDs 00200–00299, giving 100 independent simulation skies. The CMB-only arm evaluates IDs 00200–00999 except 00970, giving 799 skies. Thus calibration and evaluation IDs are disjoint, but the two evaluation arms share 100 underlying CMB simulation identities and cannot be treated as independent replications. The paired carriers additionally contain noise. No observed row, noise map, or raw map is opened by this experiment.

The matched-null exchangeability premise behind the CMB-only finite-rank interpretation does *not* hold for the paired arm: paired queries are CMB-plus-noise carriers whereas all 200 references are CMB-only. Accordingly, the paired-arm 10/201 rejection fractions describe performance of the frozen decision rule under this registered distribution shift. They are not exact matched-null p calibrations. The CMB-only arm is the matched conditional finite-pool evaluation.

For each injected query and each coordinate, the query is joined to the 200 reference rows. Both frozen reducers act on this 201-row pool. If k_j is the resulting integer local-rank numerator for coordinate j , two family combiners are applied:

$$M = \min_j k_j, \quad P = \prod_j k_j.$$

The family numerator is the number of complete-pool rows having its own M or P no larger than the query value. The product is compared as an exact integer; no χ^2 approximation or coordinate-independence assumption is introduced. A trial rejects precisely when this family numerator is at most 10. This fixes the nominal threshold $10/201 < 0.05$ before the injected results are read.

Power first averages the 32 paired orientations within each simulation sky, then averages over skies. The reported standard error is the standard error of these sky-level orientation averages, conditional on the fixed 200-row reference set. Orientations are not pseudoreplicated as independent skies. The five templates are separate alternatives, not random draws from a population; any template average in a figure is only a visual compression.

The complete execution contains 96,000 positive-amplitude injected carriers in the paired arm and 767,040 in the CMB-only arm. Both arms completed with zero typed absences. All local and family ranks remain exact integers. The unaltered $\ell = 4, 5$ blocks and proper-rotation metric were checked directly.

9.3 Null-injection behavior

Table 9 gives the rejection fractions at $A = 0$. Because the evaluation skies are finite and the 201-row family statistic is discrete, these empirical fractions need not equal 0.05. The ECDF/minimum

MES-10 combination rejects no held-out row in either arm. This is a real conservatism of that registered finite functional, not a missing result.

Table 9: Zero-injection rejection fractions. “Min” is the minimum-local- p combiner and “Prod” the exact integer-product combiner.

Family	Reducer	Combiner	Threshold	Paired	CMB-only
MES-10	Abs.-median	Min	10/201	0.0500	0.0300
MES-10	Abs.-median	Prod	10/201	0.0200	0.0501
MES-10	ECDF	Min	10/201	0.0000	0.0000
MES-10	ECDF	Prod	10/201	0.0300	0.0350
Irrep-8	Abs.-median	Min	10/201	0.0300	0.0501
Irrep-8	Abs.-median	Prod	10/201	0.0500	0.0713
Irrep-8	ECDF	Min	10/201	0.0000	0.0175
Irrep-8	ECDF	Prod	10/201	0.0800	0.0839

9.4 Power curves, crossings, and functional specificity

Table 10 compresses the five distinct alternatives by an equal template average at three amplitudes. It is not a prior-weighted or population-averaged power. The close paired/CMB-only values are a useful stability check: across every registered cell, their median absolute power difference is 0.00271, their mean difference is 0.00485, and the maximum is 0.0301 at the zero-injection MES-10 absolute-median/product cell. Shared CMB identities prevent interpreting this agreement as independent confirmation.

Table 10: Descriptive template-mean power. Each entry is paired/CMB-only.

Family	Reducer	Combiner	Meaning	$A = 2$	$A = 4$	$A = 8$
MES-10	Abs.-median	Min	five-template mean	.615/.619	.827/.830	.963/.964
MES-10	Abs.-median	Prod	five-template mean	.519/.519	.722/.720	.805/.809
MES-10	ECDF	Min	five-template mean	.000/.000	.000/.000	.000/.000
MES-10	ECDF	Prod	five-template mean	.436/.434	.709/.705	.828/.828
Irrep-8	Abs.-median	Min	five-template mean	.971/.972	1.000/1.000	1.000/1.000
Irrep-8	Abs.-median	Prod	five-template mean	.607/.607	.656/.655	.673/.675
Irrep-8	ECDF	Min	five-template mean	.110/.110	.256/.253	.288/.289
Irrep-8	ECDF	Prod	five-template mean	.558/.556	.594/.592	.579/.577

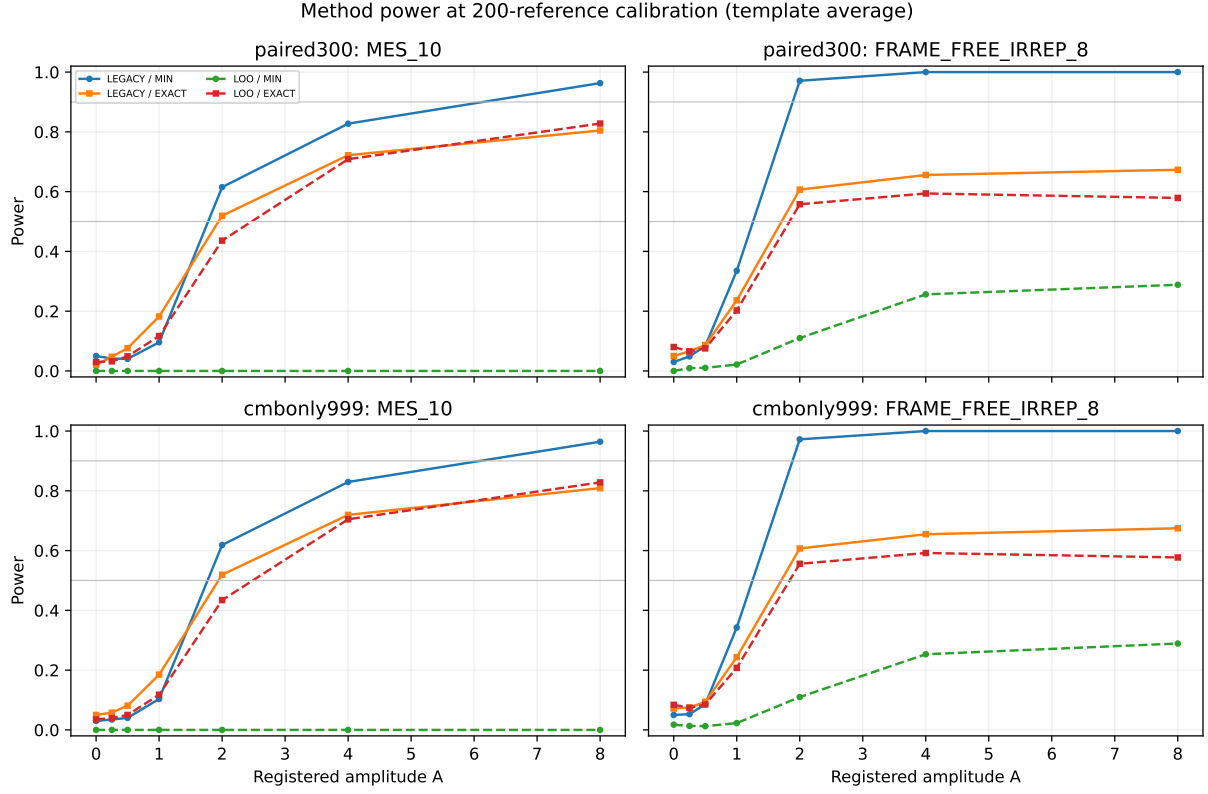


Figure 12: Registered method power versus amplitude. Positive-amplitude points are equal-template visual summaries; every template remains separately stored and scored. Grey lines mark 0.5 and 0.9. The two arms overlap in CMB identities and are not independent curves. Allowed use is conditional method- power comparison for the registered 200-reference functional only.

Template resolution is essential. For the ECDF/min-local- p irrep family, `MIX_AXIAL` is the only alternative to cross both 50% and 90% on the grid, with both crossings bracketed by $A \in [2, 4]$. Its paired/CMB-only powers are 0.4350/0.4356 at $A = 2$ and 0.9950/0.9817 at $A = 4$. At $A = 8$ the same functional reaches only 0.4106/0.4111 for `MIX_PLANAR`, about 0.022/0.023 for `Q_AXIAL`, 0.010/0.011 for `O_PLANAR`, and zero for `MIX_GENERIC`. By contrast, the absolute-median/min-local- p irrep functional reaches power one for every template at $A = 8$. The method therefore has no single template-independent “sensitivity.”

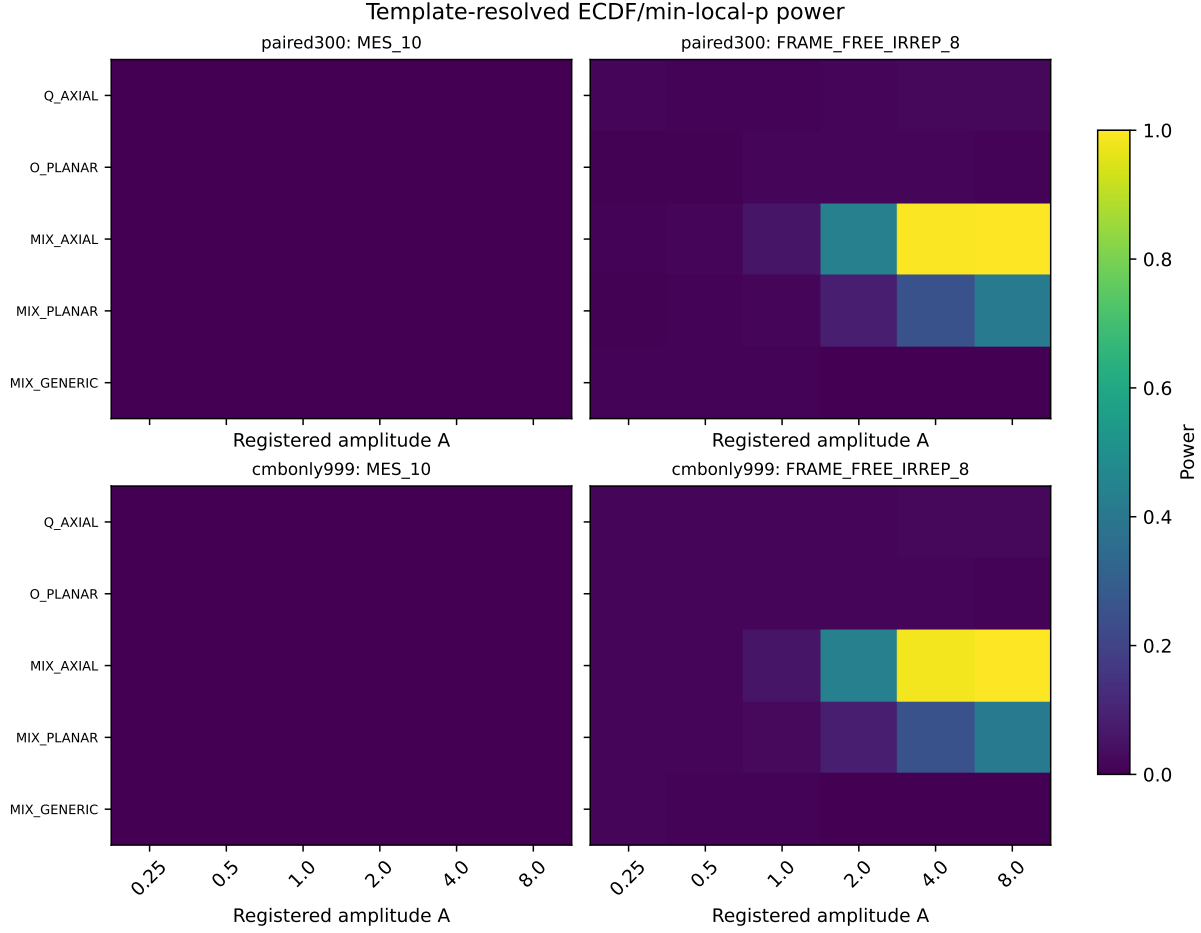


Figure 13: Template-resolved ECDF/min-local- p power. The zero MES-10 panels are the observed finite-pool saturation of this exact family combiner, not zero carrier response. The mixed-axial irrep alternative is qualitatively different from the other registered algebraic alternatives. This is a C2 conditional diagnostic, not a physical-template or observation-test result.

The exact product combiner resolves information hidden by minimum-rank ties. At $A = 8$, ECDF/product MES-10 power is approximately one for Q_AXIAL, MIX_AXIAL, and MIX_PLANAR, 0.950 for O_PLANAR, but only 0.196/0.199 for MIX_GENERIC. For the irrep family at the same amplitude it is one for the two structured mixed templates, about 0.357 for Q_AXIAL, 0.267/0.274 for O_PLANAR, and 0.270/0.253 for MIX_GENERIC. Thus the product and minimum combiners answer different, predeclared questions about distributed versus single-coordinate extremeness.

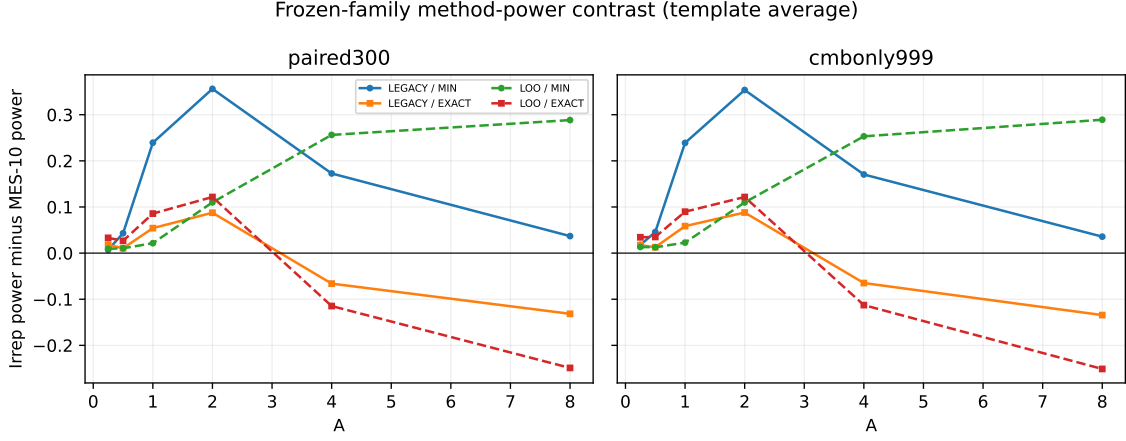


Figure 14: Template-mean irrep-8 minus MES-10 power. Positive and negative differences coexist and change with amplitude, reducer, and combiner. The figure compares observer-space statistical functionals, not physical models.

No monotonic curve was imposed: 61 of the 80 template–family–reducer–combiner curves are nondecreasing on the positive grid and 19 are not. Across all registered functionals and templates, 48 of 80 cases have at least one 50% grid crossing and 42 of 80 have at least one 90% crossing. Crossings are reported only as adjacent amplitude brackets; there is no interpolation or extrapolation and downward recrossings are retained.

The maximum absolute first-16 versus full-32 orientation power difference is 0.035625. This is a bounded quadrature-sensitivity diagnostic, not a proof of continuum Haar convergence.

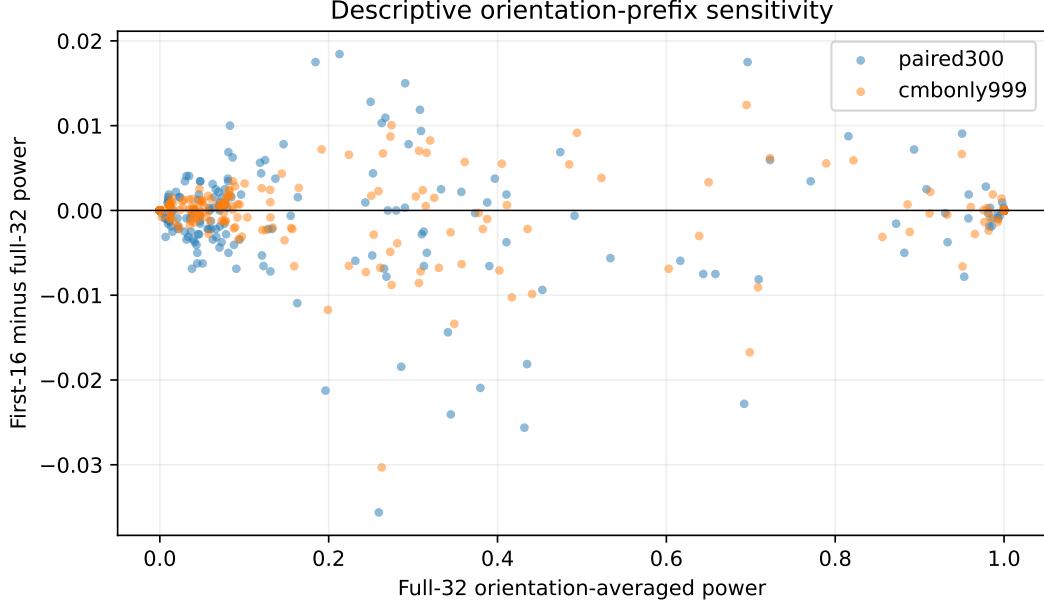


Figure 15: First-16 minus full-32 orientation-averaged power for every positive cell. Orientations are paired within a sky and do not enlarge the independent sky denominator.

9.5 Referee audit and bounded second revision

The single independent WU-008 referee audit verified the exact execution head and tree, all 13 manifested artifact hashes, and the embedded report-asset copies. An independent reconstruction from the committed rank arrays found zero mismatches among 496 power summaries and reproduced

the threshold, monotonicity, arm-difference, and orientation-sensitivity counts. Physics, finite-pool arithmetic, numerical execution, claim ownership, and the absence of physical-response promotion were otherwise consistent.

The audit nevertheless rejected the first report revision for two P1 defects: the claim-bearing result pack did not yet carry the repository’s complete machine-readable use boundary, and the four-across heatmap title collided with its panel titles at both required display widths. The one bounded repair adds a single result-pack metadata block owned by OBSSTAT, with scope, C2 conditional claim tier, allowed and forbidden uses, no-transfer status, inherited sky/mask status, covariance role, arm-specific null status, caveats, generating procedure, and exact input/candidate identities. The same block is bound into both the result and manifest and is checked on replay. The heatmap is now 2×2 with constrained layout; its data are unchanged. The two P2 clarifications are incorporated here: the paired-arm null mismatch is stated beside the decision rule, and `replay_status=MATCH` means committed- artifact integrity and summary-content replay, not reinjection of all 863,040 carriers. No rank, tail, reducer, threshold, template, tolerance, or scientific expectation changed in the repair.

10 Synthesis and scientific interpretation

10.1 What changed with the latest framework

The current methodology replaces three potential shortcuts with explicit separations.

1. **Ceiling versus state.** $\Sigma_{\max}^2$ and $W_{\max}^2$ are squared, premise-conditioned upper coordinates. They are neither the physical tensors nor estimates of their directions.
2. **Scalar versus shape.** Directional morphology comes from the measured harmonic blocks and their STF representatives, never from scalar reconstruction.
3. **Exact rank versus null fidelity.** A finite rank can be arithmetically exact relative to a frozen pool while its unconditional scientific calibration remains limited by exchangeability and simulation fidelity.
4. **Observation rank versus method power.** The observed 301- and 1000-row ranks are not assigned the power of a different 201-row reference design. The intervention experiment evaluates only its explicitly registered functionals and never uses the observation.

The scalar audit shows that the MES ceilings themselves are not locally unusual in the paired pool: both are 74/301. The smallest scalar local rank is the octupole morphology coordinate $d_{3,0}$ at 16/301. The observer- space upgrade then finds a more localized mixed-shape feature: $\hat{R}_{v0}$ and $\hat{R}_{v2}$ are 4/301 in the paired pool and 9/1000, 21/1000 in the CMB-only pool. After the predeclared family scan, the corresponding frame-free ranks are 27/301 and 61/1000 or 55/1000. This distinction between local and family ranks is essential.

The component-replacement audit adds a different sensitivity axis without altering that statistic. SMICA and NILC give the same frame-free family rank under both reducers; Commander shifts moderately; SEVEM shows the largest reducer dependence. WMAP9 ILC transports the same morphology closely, while CosmoGlobe and BeyondPlanck CMB products move both tensor shape and family rank. The fixed foreground bank then fails to show exceptional simultaneous Q/O alignment (144/301). Together these are evidence of partial, not universal, cross-product stability and evidence against one narrowly specified single-template foreground projection.

The carrier intervention adds a prospective sensitivity diagnostic. It shows strong response to some algebraic alternatives but also exposes finite- pool saturation and sharp functional specificity. The

paired and CMB-only arms closely agree cell by cell, yet are not independent because their CMB simulation identities overlap. The result therefore supports the conclusion that the recorded power curves are reproducible consequences of the frozen carrier/statistic pipeline; it does not transfer that power to a physical template, the observation test, or an unconditional null model.

10.2 What the results do not identify

The quantities Q_{ab} and O_{abc} are observer-space representations of CMB harmonics. They are not the spacetime shear tensor, vorticity tensor, anisotropic spatial curvature, or a Bianchi template. No admitted forward response connects the measured orbit coordinates to those physical states. No depth-indexed directional channel supplies the rank required by Eq. (14). Therefore no result here identifies a local boost, global tilt, physical source, or geometry family.

The present analysis also does not produce the separate diagnostic vocabulary (x, Q, Π, F, G) , a likelihood, posterior, evidence ratio, posterior predictive check, or cross-validation score. Similar symbols in the STF section—for example Q_{ab} and J_Q —must not be confused with that separate reporting layer.

10.3 Claim audit

Table 11: Admissible and inadmissible conclusions.

Statement	Status	Boundary
The registered MES ceilings obey Eqs. (5) and (6)	Supported	Explicit premises and non-negative branch
The MES transform adds independent information to C_2, C_3	Rejected	Invertibility, Eq. (10)
The scalar MES-squared family has legacy rank 98/301	Supported	Paired finite pool; absolute-median reducer
The all-two-sided ECDF companion rank is 148/301	Supported	Same rows; reducer sensitivity, not a new sky
The physics-oriented ECDF tail companion is 93/301	Supported	Same rows and cell; predeclared tail sensitivity
The paired frame-free STF family has rank 27/301 under both reducers	Supported	Conditional 301-row diagnostic
The CMB-only frame-free ranks are 61/1000 and 55/1000	Supported	Separate robustness ensemble
The PR3 component frame-free L/E ranks are SMICA 27/27, Commander 46/41, NILC 27/27, and SEVEM 81/40, all over 301	Supported	Exact observation-row replacement; unchanged paired pool and statistic
WMAP9 ILC, CosmoGlobe, and BeyondPlanck CMB are external holdouts	Supported as descriptive transport	Planck FFP10 is not asserted exchangeable with these products
The fixed 13-template foreground-bank rank is 144/301	Supported	Necessary-condition single-template Q/O projection diagnostic
The method-power study executed 96,000 and 767,040 positive injected carriers	Supported	200-reference, observation-free, map-free registered design

Statement	Status	Boundary
The paired/CMB-only cellwise power difference has median absolute value 0.00271	Supported as descriptive stability	Arms share 100 CMB identities; not independent replication
The ECDF/min-local- p MES-10 functional has zero power on every registered positive cell	Supported	Conservative 201-row family-minimum saturation; other combiners retain power
The intervention calibrates power of the original 301/1000 observation tests	Rejected	Different reference size, role graph, and no observed row
Algebraic carrier injections measure sensitivity to physical shear, vorticity, or a Bianchi solution	Rejected	No authorized physical response or physical template
All foreground explanations are ruled out	Rejected	Incomplete bank; no mixtures, nonlinear leakage, or causal source identification
The lower CMB-only fractions independently confirm an anomaly	Rejected	Same observation and family; different null pool
The observed tensors measure physical shear or vorticity	Rejected	Observer-space harmonic tensors only
Planck alone identifies local boost versus global tilt	Rejected	No independent depth response; rank deficient
A Bianchi family or physical source is identified	Rejected	No admitted physical response or native morphology atlas
The finite ranks are unconditional p -values or posteriors	Rejected	Null exchangeability and fidelity remain conditional
The registered orbit coordinates are globally complete	Unproven	Useful partial catalogue with typed degeneracies

11 Limitations and next evidence requirements

Central limitation. The strongest present result is a conditional observer-space morphology diagnostic. Increasing exactness of row identity or numerical replay cannot substitute for a missing physical response or for null-model fidelity.

1. **Null fidelity.** The paired pool represents CMB plus admitted noise; the 999-row lane represents CMB only. The component-replacement and holdout rows retain the paired Planck pool. None spans every foreground, calibration, beam, mask, or cross-instrument uncertainty.
2. **Bounded product coverage.** PR3 Commander/NILC/SEVEM and three external CMB holdouts were executed, but no compatible PR3 split was locally admitted. WMAP low-frequency, QUIJOTE, and CLASS temperature-morphology lanes remain typed exclusions for the reasons in the preceding section.
3. **Foreground scope.** The 144/301 bank diagnostic addresses one necessary condition for a simple single-template projection. It does not cover arbitrary linear mixtures, spatially varying spectra, nonlinear mapmaking leakage, or omitted components.
4. **Finite resolution and dependence.** Ranks lie on finite grids, and coordinates are dependent. Participation ratios are diagnostics, not multiplicity-free physical dimensions.
5. **Power-design scope.** The intervention fixes 200 reference carriers, five algebraic templates, 32 orientations, and two family combiners. Its empirical power is conditional on those choices. Template means are visual summaries, 19 of 80 positive-grid curves are nonmonotone, and the orientation prefix is not a continuum-convergence certificate.
6. **Premise dependence.** The principal MES numbers use the geodesic, linear almost-EGS, registered-frame, residual- $\epsilon_1 = 0$ branch. Other branches require their own recovered coefficients and assumptions.
7. **No physical response.** A future claim about shear, vorticity, local/global origin, or geometry requires a content-defined forward model, compatible conventions, rank analysis, and independent observational channels.
8. **No family identification.** A scalar rank, a sky axis, and an observer-space STF orbit are each insufficient. Family identification additionally requires a validated physical morphology atlas and null, mask, covariance, and equivalence-class gates.

The shortest scientifically meaningful next steps are therefore compatible split/null-fidelity sensitivity, explicit cross-instrument null construction, and, separately, a genuine physical response with sufficient directional and depth rank. None is silently credited as completed here.

12 Conclusion

The current MES-bound framework is now an explicit, typed chain rather than a scalar-to-physics shortcut. Low-order temperature powers yield one-way, premise-conditioned squared ceilings. Those ceilings can normalize compatible amplitudes but cannot create direction or independent information. Actual directional morphology is carried by the measured harmonic blocks and their reversible STF representatives. A physical interpretation remains a separate inverse problem whose response and rank are not yet available.

On the paired 300-row primary, the scalar MES-squared family has legacy rank 98/301 and all-two-sided ECDF rank 148/301; the physics-oriented ECDF tail companion is 93/301, and the two

MES anchors are individually 74/301. The frame-free observer-space family has 27/301 under both reducers, with local minima 4/301 in two mixed contractions. In the separate 999-row CMB-only pool, the frame-free ranks are 61/1000 and 55/1000. Under exact product replacement, the PR3 component frame-free absolute-median/ECDF ranks are SMICA 27/27, Commander 46/41, NILC 27/27, and SEVEM 81/40, all over 301. WMAP9 ILC transports the SMICA tensor shape closely, whereas CosmoGlobe and BeyondPlanck show larger shifts. A fixed 13-template foreground bank gives 144/301, so the SMICA Q/O morphology is not exceptionally aligned with a single admitted foreground template under the predeclared necessary-condition test.

The registered observation-free intervention evaluates 863,040 positive carrier injections across two overlapping evaluation arms. Their cellwise power is closely stable, while the response varies strongly with template, family, reducer, and combiner. The mixed-axial alternative is especially visible to the frame-free irrep family, whereas the ECDF/min-local- p MES-10 functional is completely conservative at denominator 201. Exact integer- product scores recover substantial response in many cells. These statements characterize the registered 200-reference method-power design only.

These results establish coordinate-, reducer-, frame-, null-ensemble-, and component-product sensitivity within precisely specified finite pools. They do not establish a detection, complete foreground validation, a physical kinematic state, a local/global origin, or a Bianchi family.

A Explicit harmonic–STF normalization

For the quadrupole define

$$A = \sqrt{\frac{5}{16\pi}}a_{20}, \quad C = \sqrt{\frac{15}{32\pi}}, \quad D = \sqrt{\frac{15}{8\pi}}.$$

The registered tensor is

$$Q = \begin{pmatrix} -A + 2C\Re a_{22} & -2C\Im a_{22} & -D\Re a_{21} \\ -2C\Im a_{22} & -A - 2C\Re a_{22} & D\Im a_{21} \\ -D\Re a_{21} & D\Im a_{21} & 2A \end{pmatrix}.$$

It is real, symmetric, and trace-free.

For the octupole let

$$E = \sqrt{\frac{7}{16\pi}}, \quad F = \sqrt{\frac{21}{64\pi}}, \quad G = \sqrt{\frac{105}{32\pi}}, \quad H = \sqrt{\frac{35}{64\pi}}.$$

Seven independent symmetric components are

$$\begin{aligned} O_{xxx} &= 2F\Re a_{31} - 2H\Re a_{33}, & O_{xxy} &= -\frac{2}{3}F\Im a_{31} + 2H\Im a_{33}, \\ O_{xxz} &= -Ea_{30} + \frac{2}{3}G\Re a_{32}, & O_{xyy} &= \frac{2}{3}F\Re a_{31} + 2H\Re a_{33}, \\ O_{xyz} &= -\frac{2}{3}G\Im a_{32}, & O_{yyy} &= -2F\Im a_{31} - 2H\Im a_{33}, \\ O_{yyz} &= -Ea_{30} - \frac{2}{3}G\Re a_{32}. \end{aligned}$$

Full symmetry supplies all permutations and trace-freedom fixes

$$O_{xzz} = -O_{xx} - O_{xyy}, \quad O_{yzz} = -O_{xxy} - O_{yyy}, \quad O_{zzz} = -O_{xxz} - O_{yyz}.$$

B Compact glossary

Term	Meaning
MES ceiling	One-way premise-conditioned upper coordinate for a squared Hubble-normalized invariant
Observer-space	Constructed from measured or simulated sky harmonics; not a physical spacetime source state
STF tensor	Symmetric trace-free Cartesian representative of a fixed angular multipole
Local rank	Observation-inclusive tail rank for one row and one coordinate
Family rank	Rank of each row's most extreme coordinate after the same scan is applied to every row
Conditional null	The specific admitted simulation pool; not an unconditional model of all uncertainty
Frame-free family	Eight proper-rotation-invariant coordinates with parity recorded explicitly and no absolute sky axis
Galactic companion	Frame-free family plus two powers relative to the Galactic pole; descriptive only
Method power	Rejection frequency for a specified injected alternative and the registered 200-reference scoring functional; not observation-test power
Typed absence	Explicitly unavailable value on a degenerate stratum; never an invented axis or silent zero
Identified set	The range of physical states compatible with a rank-deficient response, rather than an unsupported point estimate

C Evidence and identity statement

The numerical tables and fifteen figures in this report are direct products of the frozen analyses described in the text. The scalar mechanism result uses 301 rows; the paired observer-space primary uses the same 301-row role graph; the CMB-only robustness result uses 1000 rows with the stated single missing simulation identifier. The harmonic–STF residual, scalar closure, row order, family definitions, tails, and rank numerators were checked against the frozen scientific content before this synthesis. The component-replication extension opened 24 admitted observation products read-only, opened zero null maps, preserved raw-data stat identities, and replayed its committed numerical carrier content with status **MATCH**. Its result content identifier is `sha256:53ca29cac4b7c90fda5901ec70dd883cc362f9e1487ec7cfb7e4cf1cc0fc2785`. The WMAP/QUIJOTE/CLASS exclusions are part of the result, not omitted failures. The fresh review independently reproduced zero table or tensor mismatches, the 13-template bank score 0.07279657967546829 and rank 144/301, the map-free replay status, and the report's numerical statements. Its verdict is **PASS** with $P_0=0$, $P_1=0$, and two non-blocking P_2 findings; its receipt has local content identity `sha256:2de2ca13876d8222f2e68ce8b6743d83d391e07dcc68888566f98c215c09d61e`.

The method-power extension consumed only the two committed map-free carrier packages. It used one common 200-row calibration set, completed both held-out arms with zero typed absences, and recorded all 863,040 positive injected carriers in exact integer-rank arrays. The numerical registry has SHA-256 `f4374966e1710be6490e57f35ae8d725806f5488fc3595db0d3afb1ce67b5e4d`. Its independent referee envelope has SHA-256 `46f9f3e4b80f7ac941f0c6894401ab30fc6b389c0cd9da73b625dfb1b6bf2c9`. After the one bounded repair, the result/manifest claim metadata agree, both P_1 findings are closed, and committed-artifact replay is **MATCH**. The final manifest content identifier is `sha256:2e026448583006ceaba49a929d63e5ce19ad8cfb3796d30c1eb23182a5051cc1`; the reviewed terminal records $P_0=0$, $P_1=0$, `claim_promotion=false`, and no raw-map access or

mutation.

Identity is typed. Immutable scientific inputs and deterministic numerical tables require their declared exact content identities. Numerical arrays are accepted by their predeclared structural, residual, and numerical criteria. PDF serialization is packaging: a changed PDF byte stream is not by itself a changed rank or scientific result. Result validity, provenance validity, and packaging identity are therefore assessed separately. No tolerance, tail, feature family, or expected result was changed in preparing this report.

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